LOCKHEED HORIZONS



Winter 1981/82

74-90 BLDG 602 PLANT 10
R C QUIST 169747

the forward fuselage and cockpit, which had over 6,000 parts in it of high curvature, thin gages, and the canopy with its complexity. This element was tested in an oven where we could determine thermal effects and develop cockpit cooling systems.

We encountered major problems in manufacturing this test unit because the first batch of heat-treated titanium parts was extremely brittle. In fact, you could push a piece of structure off your desk and it would shatter on the floor. It was thought that we were encountering hydrogen embrittlement in our heat-treat processes. Working with our supplier, Titanium Metals Corporation, we could not prove that the problem was in fact hydrogen. It was finally resolved by throwing out our whole acid pickling setup and replacing it with an identical reproduction of what TMCA had at their mills.

We developed a complex quality control program. For every batch of ten parts or more we processed three test coupons which were subjected to the identical heat treatment of the parts in the batch. One coupon was tensile tested to failure to derive the stress-strain data. A quarter-of-an-inch cut was made in the edge of the second coupon by a sharp scissor-like cutter and it was then bent around a mandrel at the cut. If the coupon could not be bent 180° at a radius of X times the sheet thickness without breaking, it was considered to be too brittle. (The value of X is a function of the alloy used and the stress/strain value of the piece.) The third coupon was held in reserve if any reprocessing was required.

For an outfit that hates paperwork, we really deluged ourselves with it. Having made over 13 million titanium parts to date we can trace the history of all but the first few parts back to the mill pour and for about the last 10 million of them even the direction of the grain in the sheet from which the part was cut has been recorded. On large forgings, such as landing gears, we trepanned out 12 sample coupons for test before machining each part. We found out the hard way that most commercial cutting fluids accelerated stress corrosion on hot titanium so we developed our own.

Titanium is totally incompatible with chlorine, fluorine, cadmium, and similar elements. For instance, we were baffled when we found out that wing panels which we spot welded in the summer, failed early in life, but those made in the winter lasted indefinitely. We finally traced this problem to the Burbank water system which had heavily chlorinated water in the summer to prevent algae growth but not in the winter. Changing to distilled water to wash the parts solved this problem.

Our experience with cadmium came about by mechanics using cadmium-plated wrenches working on the engine installation primarily. Enough cadmium was left in contact with bolt heads which had been tightened so that when the bolts became hot (over 600°F) the bolt heads just dropped off! We had to clean out hundreds of tool boxes to remove cadmium-plated tools.

Drilling and machining high strength titanium alloys, such as B-120, required a complete research program to determine best tool cutter designs, cutting fluids, and speeds and feeds for best metal removal rates. We had particular

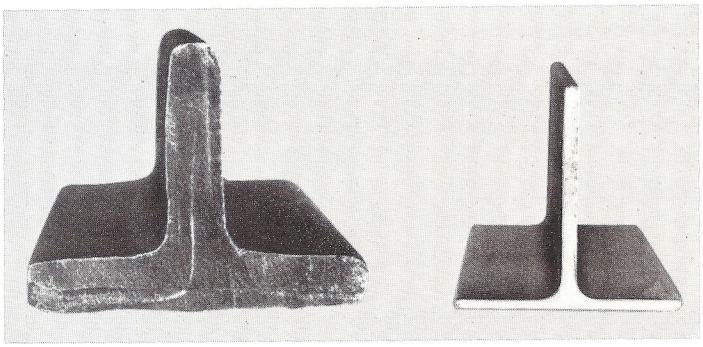


FIGURE 9 Titanium Wing Extrusion as Received from the Mill (left) and After Machining to a Finished Part (right)

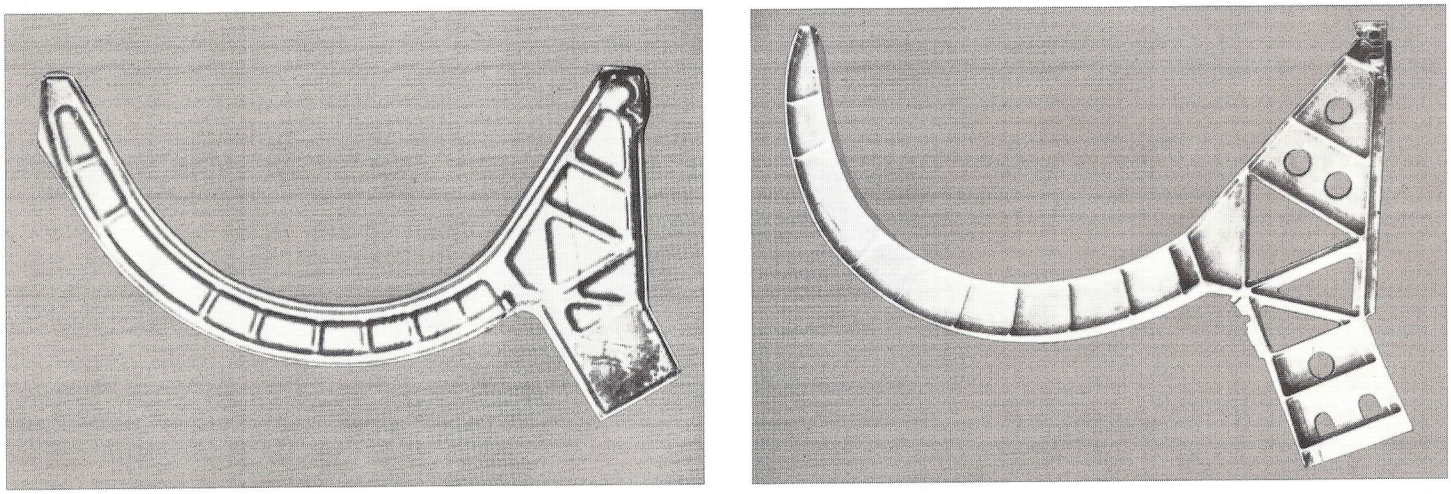


FIGURE 10 This Nacelle Ring was Originally Made from 30 parts. These Parts were Machined on Old, Outdated Profiling Machines and then Assembled into the Final Ring Segment in 487 Hours. Today, the Ring Segment is Made from a Single Forging (left) Weighing 325 pounds and is then Machined on Profilers of Advanced Design to Produce the Finished Part (right) in 150 hours.





FIGURE 11 Machine Shop with Numerical Controlled Milling Machines



FIGURE 12 Wing Fuel Tank Showing Structure and Sealant After 100 Hours Flying at Mach Numbers Over 2.6

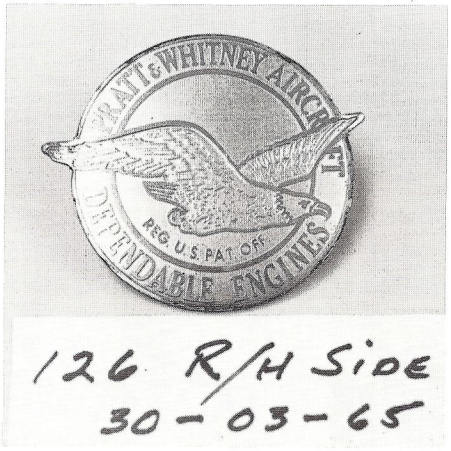


FIGURE 13 Engine Name Plate Sucked into an Engine on Ground Run Up!

trouble with wing extrusions, Figure 9, which were used by the thousands of feet. Initially, the cost of machining a foot out of the rolled mill part was $19.00 which was reduced to $11.00 after much research. At one time we were approaching the ability at our vendor's plants to roll parts to net dimensions, but the final achievement of this required a $30,000,000 new facility which was not built.

Wyman Gordon was given $1,000,000 for a research program to learn how to forge the main nacelle rings, Figure 10, on a 50,000-ton press which was successful. Combining their advances with our research on numerical controls of machining and special tools and fluids, we were able to save $19,000,000 on the production program.

To prevent parts from going undergage while in the acid bath, we set up a new series of metal gages two thousandths of an inch thicker than the standard gages and solved this problem. When we built the first Blackbird, a high-speed drill could drill 17 holes before it was ruined. By the end of the program we had developed drills that could drill 100 holes and then be resharpened successfully.

Our overall research on titanium usage was summarized in reports which we furnished not only to the Air Force but also to our vendors who machined over half of our machined parts for the program; Lockheed's machine shop for this program is shown in Figure 11. To use titanium efficiently required an ongoing training program for thousands of people—both ours in manufacturing and in the Air Force in service.

Throughout this and other programs, it has been crystal clear to me that our country needs a 250,000-ton

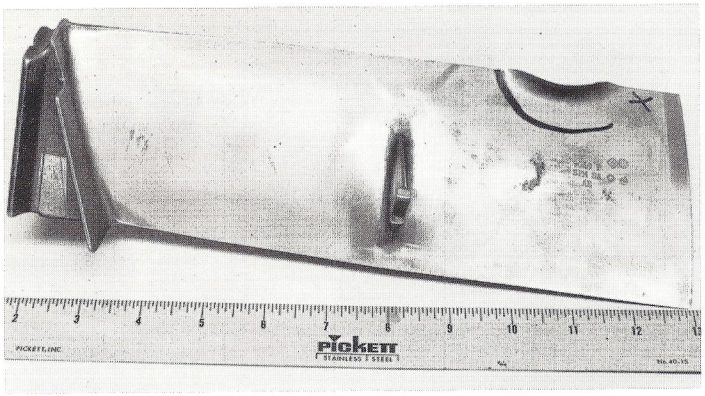
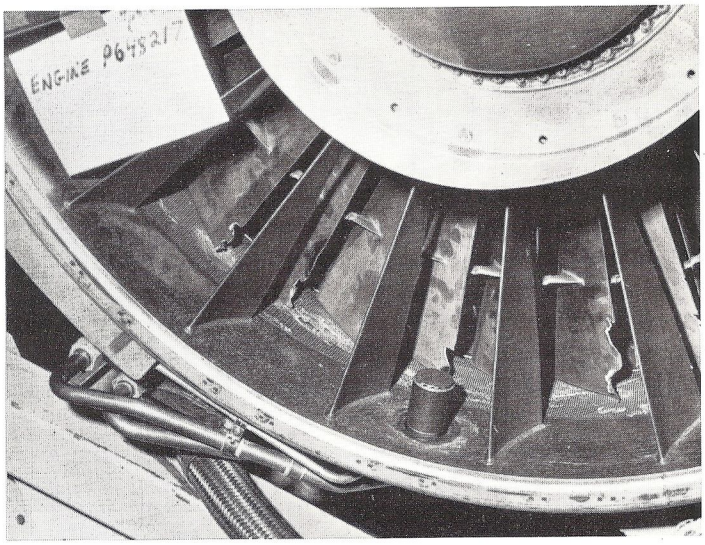


FIGURE 14 First-Stage Compressor Blade After Hitting Inspector's Flashlight—Total Engine Damage $250,000



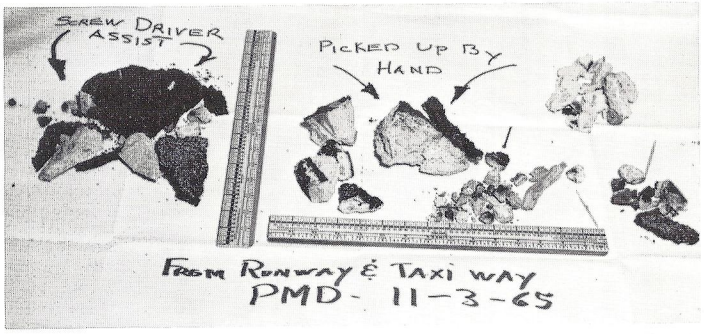


FIGURE 15 Taxiway/Runway Debris Which Could be Sucked into Engines on Take-off

metal forming press—five times as large as our biggest one available today. When we have to machine away 90 percent of our rough forgings today both in titanium (SR-71 nacelle rings and landing gears) and aluminum (C-5 fuselage side rings) it seems that we are nationally very stupid! My best and continuing efforts to solve this problem have been defeated for many years. Incidentally, the USSR has been much smarter in this field in that they have more and larger forging presses than we do.

FLUID SYSTEMS

Very difficult problems were encountered with the use of fuel tank sealants, Figure 12, and hydraulic oil. We worked for years developing both of these, drawing as much on other industrial and chemical companies as they were willing to devote to a very limited market. We were finally able to produce a sealant which does a reasonable job over a temperature range of minus 90°F to over 600°F. Our experience with hydraulic oil started out on a comical situation. I saw ads in technical journals for a "material to be used to operate up to 900°F in service." I contacted the producer who agreed to send me some for testing. Imagine my surprise when the material arrived in a large canvas bag. It was a white powder at room temperature that you certainly wouldn't put in a hydraulic system. If you did, one would have to thaw out all the lines and other elements with a blow torch! We did finally get a petroleum-based oil developed at Penn State University to which we had to add several other chemicals to maintain its lubricity at high temperatures. It originally cost $130 per gallon so absolutely no leaks could be tolerated.

Rubber O-rings could not be used at high temperatures so a complete line of steel rings was provided which have worked very well. Titanium pistons working in titanium cylinders tended to gall and seize until chemical coatings were invented which solved the problem.

THE FLIGHT TEST PHASE

The first flight of the A-12 took place 26 April 1962 or thirty months after we were given a limited go-ahead on 1 September 1959. We had to fly with Pratt & Whitney J75 engines until the J58 engine became available in January 1963. Then our problems really began!

The first one was concerned with foreign object damage (FOD) to the engines—a particular problem with the powerful J58 and the tortuous flow path through the complicated nacelle structure. Small nuts, bolts, and metal scraps, Figure 13, not removed from the nacelles during construction could be sucked into the engines on starting with devastating results. Figure 14 shows damage to the first-stage compressor blades from an inspector's flashlight used to search for such foreign objects. Engine damage—$250,000! Besides objects of the above type, the engine would suck in rocks, asphalt pieces, etc. from the taxi-ways and runways, Figure 15. An intensive campaign to control FOD at all stages of construction and operation—involving a shake test of the forward nacelle at the factory, the use of screens, and runway sweeping with double inspections prior to any engine running—brought FOD under reasonable control.

The hardest problem encountered in flight was the development of the nacelle air inlet control. It was necessary to throw out the initial pneumatic design after millions of dollars had been spent on it and go to a design using electronic controls instead. This was very hard to do because several ele-

ments of the system were exposed to ram-air temperatures over 800°F and terrific vibration during an inlet duct stall. This problem and one dealing with aircraft acceleration between Mach numbers of 0.95 to 2.0 are too complex to deal with in this paper.

Initially, air temperature variations along a given true altitude would cause the Blackbird to wander up and down over several thousand feet in its flight path. Improved autopilots and engine controls have eliminated this problem.

There are no other airplanes flying at our cruising altitude except an occasional U-2 but we were very scared by encountering weather balloons sent up by the FAA. If we were to hit the instrumentation package while cruising at over 3,000 feet per second, the impact could be deadly!

Flight planning had to be done very carefully because of sonic boom problems. We received complaints from many sources. One such stated that his mules on a pack-train wanted to jump off the cliff trail when they were "boomed." Another complained that fishing stopped in lakes in Yellowstone Park if a boom occurred because the fish went down to the bottom for hours. I had my own complaint when one of my military friends boomed my ranch and broke a $450 plate glass window. I got no sympathy on this, however.

OPERATIONAL COMMENTS

The SR-71 first flew 23 December 1964. It was in service with the Strategic Air Command a year later.

In-flight refueling from KC-135s turned out to be very routine. Over eighteen thousand such refuelings have been made to date by all versions of the Blackbirds and they have exceeded Mach 3 over 11,000 times.

The SR-71 has flown from New York to London in 1 hour 55 minutes then returned nonstop to Beale Air Force Base, including a London/Los Angeles time of 3 hours 48 minutes.

It has also flown over 15,000 miles with refueling to demonstrate its truly global range. It is by far the world's fastest, highest flying airplane in service. I expect it to be so for a long time to come.



CLARENCE L. (KELLY) JOHNSON is serving as senior advisor to corporate management and the firm's advanced development projects (Skunk Works). He retired from his post as senior vice president of the corporation in January 1975 and from the board of directors in May 1980.

Johnson joined Lockheed in 1933 as a tool designer. After assignments as flight test engineer, stress analyst, aerodynamicist, weight engineer, and wind tunnel engineer, he became chief research engineer in 1938. In 1952, Johnson was named chief engineer at Lockheed's Burbank, California plant, now the Lockheed-California Company. When the office of corporate vice president—research and development was established in 1956, he was chosen for the post. He became vice president—advanced development projects in 1958, a member of the board of directors in 1964 and a senior vice president of the corporation in 1969.

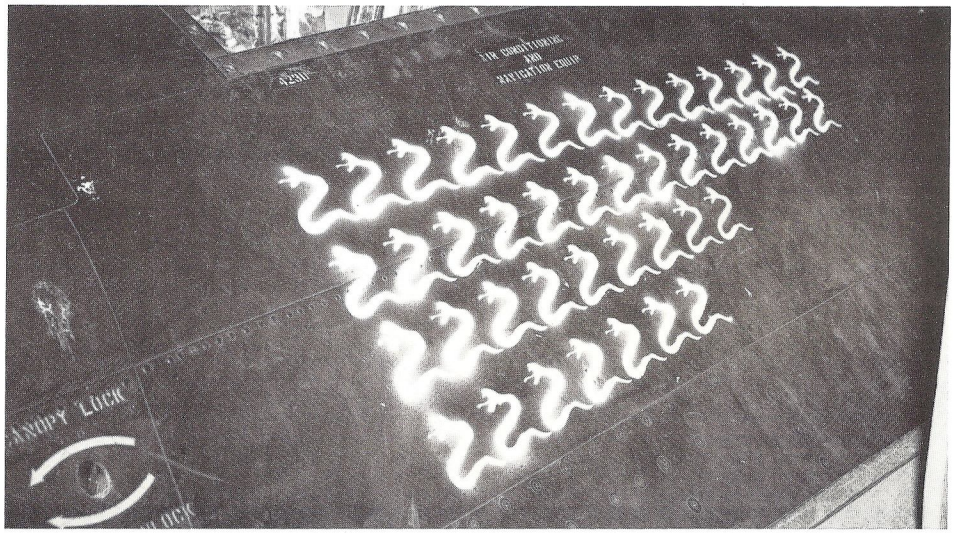
Johnson has played a leading role in the design of 40 world renowned aircraft—among them the F-80, America's first production jet; the high altitude U-2; the double-sonic F-104 Starfighter; and the spectacular 2000-m.p.h. YF-12A and SR-71.

A native of Michigan, Johnson was born in Ishpeming on February 27, 1910. He later moved to Flint, was graduated from Flint Junior College, and completed his education at the University of Michigan, where he received his bachelor of science degree in 1932 and his master of science degree in aeronautical engineering in 1933.

Many honors have come to him for his unique contributions to aerospace development through the years, and to the defense of the United States and the Free World. Among these, Mr. Johnson has won the Collier Trophy twice and has also received two Theodore von Karman and two Sylvanus Albert Reed Awards. In 1964, President Lyndon B. Johnson presented to him the Medal of Freedom, the highest civil honor the President can bestow. He was elected to the Aviation Hall of Fame in 1974 and is the 1981 recipient of the Daniel Guggenheim Award.

Johnson is an honorary fellow of the American Institute of Aeronautics and Astronautics, a fellow of the Royal Aeronautical Society, and a member of the Society of Automotive Engineers and Tau Beta Pi and Sigma Xi engineering fraternities.

Mr. Johnson is married to the former Nancy Powers and they reside in Encino, California.



Southeast Asia Combat Missions for One SR-71—June 1971